



What Should Be in the Datasets

For the AI/ML models to actually work, each data category needs specific fields and structure, not just raw feeds. Here's what each dataset should contain.

1. Rainfall Dataset (IMD)

- Station/grid ID with latitude, longitude, and elevation
- Timestamp (hourly/daily granularity minimum, sub-hourly ideal for intensity-duration analysis)
- Rainfall amount (mm), plus 1-hr, 3-hr, 24-hr, and 72-hr cumulative totals — since landslides correlate strongly with cumulative antecedent rainfall, not just single-day totals
- Forecast rainfall (short-term, 1–7 day) with confidence/probability bands
- Historical rainfall-intensity-duration thresholds specific to each district, since Wayanad's failure showed generic thresholds don't work locally^[1]

2. Soil Moisture / Deformation Dataset (Sentinel-1 InSAR)

- Grid cell ID with coordinates and acquisition date
- Radar backscatter/coherence values (to flag low-confidence zones from vegetation decorrelation)^[2]
- Displacement rate (mm/year or mm/revisit-cycle) with direction (line-of-sight)
- Coherence quality flag — critical, since this tells the model when InSAR data should be trusted versus ignored in a given cell
- Revisit date/cycle metadata, so the fusion engine knows how "stale" a reading is

3. Vegetation Stress Dataset (NDVI, Sentinel-2/Landsat)

- Grid cell ID, coordinates, acquisition date
- NDVI value and NDVI anomaly (deviation from historical seasonal baseline for that cell)
- Cloud-cover flag (NDVI readings under heavy cloud should be excluded or down-weighted)

4. Terrain Dataset (DEM — SRTM/Cartosat, ISRO Bhuvan)

- Elevation (meters), slope angle (degrees), aspect (direction)
- Curvature (planform/profile) — affects water flow concentration
- Land use/land cover classification (forest, cultivated, built-up, bare soil, quarry/hill-cut areas)

- Distance to nearest road, stream, and settlement
- Geology/lithology type where available (rock type, weathering grade)
- Soil thickness/depth-to-bedrock where obtainable — flagged as a known data gap in India, but should be actively sourced or estimated wherever possible^[3]

5. Historical Landslide Records (NRSC/GSI inventory)

- Event ID, date, location (point or polygon)
- Landslide type (debris flow, rockfall, slump, etc.), size/volume estimate
- Trigger cause (rainfall-induced, human-induced/hill-cutting, seismic)
- Casualties/infrastructure damage recorded
- Source reliability flag (verified by GSI vs. media-reported)

6. Road & Infrastructure Dataset (OSM + PWD/NHAI)

- Road segment ID, geometry (line), road class (national highway, state highway, village road)
- Nearby villages/settlements with population estimates
- Critical infrastructure points (schools, hospitals, government buildings) with coordinates
- Known chokepoints/single-access-road villages, since these are highest priority during blockage events

7. Community Rainfall Dataset (recommended addition)

- Gauge ID, exact GPS location, maintainer/volunteer contact
- Manual/automated reading timestamp and rainfall value
- Gauge calibration/maintenance log (to flag unreliable gauges)
- This directly mirrors the Wayanad community rain-gauge model, which fills the hyperlocal gap that IMD and satellite data structurally miss^[4]

Cross-Cutting Requirements for All Datasets

- Every record needs a **consistent zone ID** matching the system's fixed grid resolution (recommended: ~25 sq km, further subdivided by slope, matching the proven Wayanad approach) so all data layers can be spatially joined.^[4]
- Every record needs a **data quality/confidence flag**, since mixing high-confidence and low-confidence inputs silently (without flagging) is what causes false alarms and missed events.
- **Timestamped versioning** on every dataset — required for the plan's audit-logging feature, so any prediction or alert can be traced back to exactly which data snapshot triggered it.

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1. <https://www.newindianexpress.com/states/kerala/2024/Aug/03/reliable-rainfall-data-must-for-accurate-landslide-warning-gsi>

2. <https://www.tandfonline.com/doi/full/10.1080/19475705.2026.2672792>
3. <https://www.hindustantimes.com/india-news/wayanad-tragedy-how-kerala-can-improve-its-early-warning-systems-for-landslides-101722510782490.html>
4. <https://thebetterindia.com/changemakers/wayanad-rain-gauge-network-landslide-warning-kerala-12145591>